

MATH 180 Strategies of Problem Solving

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Outline

- 1 SoPS Lecture 12
 - Starter problem
 - Strategies of Problem Solving
 - Wrapping up

Review

Last time

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- mathematics as a creative endeavor
- problem solving strategies
- creativity in problem solving

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An old problem

Problem

What is the sum of the numbers

$$1 + 2 + 3 + 4 + \cdots + 99 + 100?$$

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Theorem

For any positive integer n ,

$$1 + 2 + 3 + 4 + \cdots + (n - 1) + n = \frac{n(n + 1)}{2}.$$

2024 IMO Shortlist

Let $\lfloor x \rfloor$ be the **floor function**, ie. what we get by rounding x down to the nearest integer. So for example

$$\lfloor 2.6 \rfloor = 2, \quad \lfloor 3 \rfloor = 3, \quad \lfloor -7.1 \rfloor = -8.$$

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For which real numbers x is the integer

$$\lfloor x \rfloor + \lfloor 2x \rfloor + \lfloor 3x \rfloor + \cdots + \lfloor nx \rfloor$$

divisible by n for all positive integers n ?

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Discussion

Question

What strategies of problem solving did you use on the previous problem?

Putnam Problem

Problem

A grasshopper starts at the origin in the coordinate plane and makes a sequence of hops. Each hop has length 5, and after each hop the grasshopper is at a point whose coordinates are both integers; thus, there are 12 possible locations for the grasshopper after the first hop. What is the smallest number of hops needed for the grasshopper to reach the point (2021, 2021)

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Final thoughts

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Reflection

- Look up an example of a Putnam Exam problem that looks interesting. What is the problem statement? What is hard about it?
- Write down some ideas about how you might begin to approach solving the problem you mentioned.

Submit your responses by **Tonight!**